# FEASIBILITY STUDY

## AI-Powered Smart Tourism Assistant Platform with Personalized Trip Planning

CS3202 – Software Engineering Project

Project ID: 14

Group: 10

Mentor: Isuru Ariyarathne


**Prepared By**

| | |
|---|---|
| Babijana Jegarashan | 230304R |
| Kajatheepan Perinpanathan | 230310G |
| Jitharsanan Thiruketheeswaran | 230305V |


July 12, 2026

# Contents

**Contribution Table** **iii**

**1 Introduction** **1**

1.1 Overview of the Project . . . . . 1

1.2 Objectives of the Project . . . . . 1

1.3 Need for the Project . . . . . 2

1.4 Overview of Existing Systems and Technologies . . . . . 2

1.4.1 Existing Similar Systems . . . . . 2

1.5 Scope of the Project . . . . . 3

1.6 Deliverables . . . . . 4

**2 Feasibility Study** **5**

2.1 Financial Feasibility . . . . . 5

2.1.1 Cost Analysis . . . . . 5

2.1.2 Estimated Cost Summary . . . . . 6

2.2 Technical Feasibility . . . . . 6

2.2.1 Technical Requirements . . . . . 6

2.2.2 Evaluation of Technology Alternatives . . . . . 7

2.2.3 Technology Comparison Summary . . . . . 8

2.2.4 Technology Integration . . . . . 9

2.2.5 Technical Challenges . . . . . 9

2.3 Resource and Time Feasibility . . . . . 9

2.4 Risk Feasibility . . . . . 10

2.5 Social and Legal Feasibility . . . . . 11

2.5.1 Social Feasibility . . . . . 11

2.5.2 Legal Feasibility . . . . . 12

**3 Considerations** **13**

3.1 Performance . . . . . 13

3.2 Security . . . . . 13

3.3 Usability . . . . . 13

3.4 Maintainability . . . . . 14

3.5 Scalability . . . . . 14

# Contribution Table

Table 1: Team Member Contributions

| **Team Member** | **Contributions** |
|---|---|
| 230304R | 2.3 Resource and Time Feasibility |
| | 2.4 Risk Feasibility |
| 230310G | 2.2 Technical Feasibility |
| | 2.5 Social and Legal Feasibility |
| 230305V | 2.1 Financial Feasibility |
| | 3 Considerations |

# 1 Introduction

## 1.1 Overview of the Project

The proposed project aims to develop an AI-powered smart tourism assistant platform that simplifies and enhances the travel planning process through a centralized mobile and web application.

Currently, tourists need to use multiple applications and websites for accommoda- tion, restaurants, attractions, weather information, and local events. This makes travel planning fragmented, time-consuming, and inefficient.

To address this issue, the proposed system will provide a single integrated platform where users can interact with an AI-powered assistant through a conversational interface and receive personalized travel recommendations and optimized itineraries based on their preferences, budget, travel duration, and interests.

The system will include a mobile application, web application, backend services, and an administrative dashboard for managing tourism-related data such as hotels, restau- rants, attractions, and local events.

The proposed system will accept various user inputs including travel preferences and interests, budget constraints, travel duration, destination or district, number of travellers, and preferred travel style. Based on these inputs, the system will generate personalized itineraries and provide attraction recommendations, hotel and restaurant suggestions, budget estimations, nearby attraction information, and optimized travel schedules tailored to the user's requirements.

## 1.2 Objectives of the Project

- Design and implement an AI-powered smart tourism assistant platform.
- Provide personalized trip planning and recommendation services.
- Develop a conversational interface for travel assistance.
- Automate itinerary generation based on user preferences and constraints.
- Provide budget planning and optimization for travellers.

- Develop an administrative dashboard for tourism content management.
- Deliver both mobile and web applications for improved accessibility.

## 1.3 Need for the Project

Tourists often rely on multiple applications and websites to obtain information about destinations, hotels, restaurants, attractions, weather conditions, and local events. This fragmented approach makes travel planning time-consuming and inconvenient, particularly for users who are unfamiliar with a destination.

The proposed smart tourism assistant addresses these challenges by providing a centralized platform that generates personalized recommendations and optimized itineraries through an AI-powered conversational interface. By integrating tourism data, budget planning, and route optimization into a single system, the platform aims to improve convenience, reduce planning effort, and enhance the overall travel experience.

## 1.4 Overview of Existing Systems and Technologies

### 1.4.1 Existing Similar Systems

- Mindtrip: An AI-powered travel planning platform that generates personalized itineraries through a conversational interface. It integrates interactive maps and allows users to organize their trips in a single platform.
- Layla: An AI travel assistant that generates travel itineraries and helps users find hotels and flights. The system redirects users to external booking platforms to complete reservations .
- Google Travel: A travel planning platform that helps users organize destinations, accommodation, flights, and travel reservations in one place. It also provides destination information and trip management features .
- Wanderlog: A trip planning and itinerary management platform that allows users to organize travel plans, collaborate with travel companions, and manage reservations across multiple devices.

Although existing travel planning platforms provide useful features, several limitations still exist:

- Most platforms depend heavily on external booking services and focus mainly on hotels and transportation rather than complete trip planning.
- Existing systems often rely on publicly available information and may not provide curated or locally verified tourism data.
- Many platforms provide itinerary suggestions but require users to manually adjust plans to satisfy practical constraints such as time, budget, and travel distance.

The proposed system addresses these limitations by providing a centralized AI-powered tourism platform that combines personalized recommendations, budget planning, itinerary generation, and route optimization within a single application. The system will utilize curated tourism data and generate travel plans based on user preferences, budget, travel duration, and interests.

## 1.5 Scope of the Project

The proposed system aims to develop an AI-powered smart tourism assistant platform that supports personalized trip planning through web and mobile applications. The system will recommend attractions, hotels, restaurants, nearby places, and travel schedules based on user preferences and budget.

### Traveler / General User

Users can register and log into the system, enter travel details such as destination, budget, travel duration, interests, and travel style, and interact with the AI assistant through a conversational interface. The system will generate personalized itineraries and recommend suitable attractions, hotels, and restaurants. Users will also be able to view nearby places and manage their travel plans.

### Administrator

The administrator will manage tourism-related information such as attractions, hotels, restaurants, and local events through an admin dashboard. The administrator will also monitor system usage and reports.

### AI Planning Module

The AI planning module will analyze user requirements and generate personalized recommendations and travel itineraries based on budget, interests, travel duration, and user preferences.

## 1.6 Deliverables

- Mobile application.
- Web application.
- AI-powered itinerary generation and recommendation engine.
- Administrative dashboard.
- Centralized tourism database.
- RESTful APIs.

# 2 Feasibility Study

## 2.1 Financial Feasibility

### 2.1.1 Cost Analysis

Financial feasibility evaluates whether the proposed AI-powered Smart Tourism Assistant Platform can be developed and maintained within a reasonable cost. Since the system is planned as an academic software project and prototype, the development approach focuses on minimizing cost by using open source technologies, free development tools, and controlled usage of external services.

The proposed system will use React.js and Tailwind CSS for the web frontend,Python and FastAPI for backend development, PostgreSQL as the database, SQLAlchemy as the ORM, JWT for authentication,GitHub for version control, and Docker for containerization. These technologies are either open source or freelly available for development purposes.Therefore, the main software development cost can be kept low.

The system also requires external services such as the Gemini API, Google Maps Platform, Google Places API, and Weather API to support AI based trip planning, route visualization, nearby attraction suggestions, and weather updates. These services may introduce usage based costs depending on the number of API requests. However,during the prototype and academic development stage, the system can be designed to operate with limited API usage, free tier access where available, and optimized request handling.

From a hardware perspective,the project does not require expensive physical infrastructure. Development can be carried out using standard personal computers or laptops. For deployment and testing, the system can initially run on a local development environment or a low cost cloud or server environment. This makes the project financially practical for the development team.

The administrative dashboard will also reduce long term data management costs because tourism related information such as attractions, hotels, restaurants and local events can be managed directly by authorized administrators without requiring database level technical changes.

The financial benefits of the system include reducing the time and effort required for travel planning, improving user convenience, and providing a centralized platform for tourism

related information. Although the system may not generate direct income during the academic prototype stage, it has potential future value through partnerships with hotels, restaurants, tour operators, travel agencies, and tourism service providers.

Overall, the proposed system is financially feasible because it can be developed using mostly free and open source technologies, does not require expensive hardware, and can control external API costs through limited and optimized usage.

### 2.1.2 Estimated Cost Summary

Table 2.1: Estimated Cost Summary

| **Cost Item** | **Estimated Cost Level** | **Description** |
|---|---|---|
| React.js, Tailwind CSS, FastAPI, PostgreSQL, SQLAlchemy | LKR 0 | Open source technologies used for system development. |
| GitHub and Docker | LKR 0 | Used for version control, collaboration, and deployment support. |
| Gemini API | LKR 0 for prototype usage | Cost depends on usage volume and selected API plan. |
| Google Maps Platform / Google Places API | LKR 0 for limited prototype usage | Cost depends on map, route, and place request volume. |
| Weather API | LKR 0 for basic prototype usage | Free or limited usage plans may be used during prototype development. |
| Hosting / Deployment | LKR 0 initially; approximately LKR 1,700–6,700 per month if paid hosting is required later | Initial testing can use local hosting or a low cost cloud/server environment. |
| Hardware | LKR 0 additional cost | Standard laptops or computers are sufficient for development and testing. |

## 2.2 Technical Feasibility

### 2.2.1 Technical Requirements

The proposed AI-powered Smart Tourism Assistant Platform requires a combination of technologies to support its core functionalities, including AI-assisted itinerary generation,

user authentication, tourism data management, and location-based services. The system also requires a responsive frontend, a reliable backend, and a database capable of securely managing user information and travel-related data.

In addition to that, the platform must integrate with AI and mapping services to provide personalized recommendations, route information, and nearby attractions. To ensure long-term usability, the selected technologies should also support scalability, maintainability, efficient integration, and future enhancements.

Based on these technical requirements, the proposed technologies are evaluated in the following section.

### 2.2.2 Evaluation of Technology Alternatives

To identify the most suitable technology stack for the proposed AI-powered Smart Tourism Assistant Platform, several alternatives were evaluated based on their performance, scalability, community support, ease of integration, and maintainability. The selected technologies provide the best balance between functionality, development efficiency, and future extensibility.

#### Frontend

The frontend requires a responsive and interactive user interface for trip planning, itinerary visualization, and user management. React.js, Angular, and Vue.js were considered. React.js was selected because of its component-based architecture, extensive ecosystem, and strong community support. Compared to Angular, React is more lightweight and flexible, while Vue.js has a relatively smaller ecosystem. These advantages make React.js well suited for developing a scalable and maintainable user interface.

#### Backend

The backend is responsible for handling business logic, processing user requests, and integrating with AI and external services. FastAPI, Django, Express.js, and Spring Boot were evaluated. FastAPI was selected due to its high performance, asynchronous request handling, automatic API documentation, and seamless integration with Python-based libraries. These capabilities make it particularly suitable for an AI-powered application while maintaining efficient API performance.

#### Database

The system requires reliable storage for user accounts, travel itineraries, attractions, hotels, restaurants, and other tourism-related information. PostgreSQL, MySQL, and MongoDB were considered. PostgreSQL was selected because it offers strong support for relational data, ensures data integrity through ACID compliance, and provides advanced querying capabilities. These features make it a reliable choice for managing structured tourism data.

#### Artificial Intelligence

The proposed platform depends on a large language model to generate personalized travel itineraries and provide conversational assistance. Gemini, OpenAI, Claude, and locally hosted language models were evaluated. Gemini API was selected because it provides strong reasoning capabilities and is readily accessible through the available student plan, making it a practical and cost effective choice for the proposed system.

#### Location Based Services

Location based services are essential for route planning, attraction discovery, and map visualization. Google Maps Platform, OpenStreetMap, and Mapbox were evaluated. Google Maps Platform was selected because it provides comprehensive mapping services, accurate place information, routing capabilities, and well documented APIs. These features effectively support the location-based functionalities required by the platform.

### 2.2.3 Technology Comparison Summary

Table 2.2: Technology Comparison Summary

| **Component** | **Alternatives Evaluated** | **Selected Technology** |
|---|---|---|
| Frontend | React.js, Angular, Vue.js | React.js |
| Backend | FastAPI, Django, Express.js, Spring Boot | FastAPI |
| Database | PostgreSQL, MySQL, MongoDB | PostgreSQL |
| AI Platform | Gemini, OpenAI, Claude, Local LLM | Gemini API |
| Maps | Google Maps Platform, OpenStreetMap, Mapbox | Google Maps Platform |

### 2.2.4 Technology Integration

The selected technologies can be effectively integrated to support the proposed AI-powered Smart Tourism Assistant Platform. The React.js frontend communicates with the FastAPI backend through RESTful APIs, while the backend uses SQLAlchemy ORM to interact with the PostgreSQL database for storing and retrieving application data. The backend also integrates with external services, including the Gemini API for AI-powered itinerary generation, the Google Maps Platform for location-based features, and the Weather API for providing weather information.

The proposed system follows a modular architecture, allowing each component to operate independently while working together as a complete solution. This improves maintainability, simplifies future enhancements, and ensures that the selected technology stack can reliably support the system's functional requirements.

### 2.2.5 Technical Challenges

Although the selected technology stack is suitable for the proposed system, several technical challenges may arise during development and deployment. The platform relies on external services such as the Gemini API, Google Maps Platform, and Weather API, which may be affected by usage limits, temporary service interruptions, or changes in pricing policies. In addition, AI-generated responses may occasionally produce inaccurate or incomplete travel recommendations.

The system may also experience increased response times as the number of users and API requests grows. These challenges can be mitigated through efficient API usage, proper error handling, caching frequently accessed data, and validating AI-generated information before presenting it to users where necessary.

Despite these challenges, no significant technical barriers have been identified that would prevent the successful implementation of the proposed system.

## 2.3 Resource and Time Feasibility

Resource and time feasibility evaluates whether the proposed system can be successfully developed using the required resources within the planned development schedule. This analysis considers the availability of software technologies, external services, development resources, and the implementation approach necessary to deliver the proposed system.

The proposed AI-powered smart tourism assistant platform requires a combination of

frontend, backend, database, artificial intelligence, and location-based technologies to provide its core functionalities. The selected technology stack supports conversational trip planning, personalized itinerary generation, tourism data management, authentication, and location-aware services. External services such as the Gemini API, Google Maps Platform, Google Places API, and Weather API provide capabilities that would otherwise require significant development effort, making the implementation more practical while reducing development complexity.

From a resource perspective, the required software frameworks, development tools, and external services are readily available and well supported. The proposed system also adopts a modular architecture that separates the frontend, backend, AI services, database, and administrative dashboard into independent components. This architecture enables different modules to be developed, tested, and integrated concurrently, improving development efficiency and simplifying future maintenance.

From a time perspective, the modular design and phased development approach support the completion of the project within the allocated development period.The project timeline has been planned through a separate Project Schedule (Gantt Chart), which outlines the major development phases, milestones, and task dependencies. Independent development of major system components reduces integration complexity and minimizes delays during implementation and testing. Based on the availability of the required technological resources and the proposed implementation strategy, the project is considered feasible in terms of both resource availability and development time

## 2.4 Risk Feasibility

Risk feasibility evaluates the uncertainties that may affect the successful development, deployment, and operation of the proposed system. Identifying potential risks at an early stage allows appropriate mitigation strategies to be incorporated into the system design, reducing their impact on project success and long-term operation.

One of the primary risks associated with the proposed system is its dependence on external services such as the Gemini API, Google Maps Platform, Google Places API, and Weather API. These services are essential for functionalities including conversational trip planning, route optimization, nearby attraction recommendations, and real-time weather updates. Any service interruption, API usage limitation, or changes to pricing policies may temporarily affect these features. To minimize this risk, the system will implement proper exception handling, optimize API requests to reduce unnecessary usage, and cache frequently accessed information where appropriate. These measures help maintain acceptable system functionality even when external services experience temporary disruptions.

Another significant risk is the quality of tourism-related information stored in the system. Personalized recommendations and itinerary generation depend on accurate information about attractions, hotels, restaurants, and local events. Outdated or incorrect information could lead to poor recommendations and reduce user confidence in the platform. This risk will be addressed through the administrative dashboard, which enables authorized administrators to regularly review, update, and verify tourism data before it is presented to users.

System performance is another important consideration. As the number of users increases, higher volumes of API requests and database transactions may increase response times and reduce overall system performance. To address this risk, the system will adopt a scalable architecture, optimize database queries, and utilize Docker-based containerization to support future expansion and efficient resource management.

The proposed platform also integrates several independent components, including the frontend application, backend services, database, AI services, and external APIs. Integration issues between these components may result in communication failures or unexpected system behavior during development. This risk can be minimized by adopting a modular system architecture, following standardized REST API communication, and performing continuous integration and system testing throughout the development process.

Although the proposed system involves several technical and operational risks, the identified mitigation strategies significantly reduce their potential impact. Therefore, the project is considered feasible from a risk perspective.

## 2.5 Social and Legal Feasibility

### 2.5.1 Social Feasibility

The proposed AI-powered Smart Tourism Assistant Platform is considered socially feasible as it addresses common challenges faced by travelers when planning trips. By providing personalized itineraries, location-based recommendations, and travel-related information through a single platform, the system reduces the time and effort required to plan a trip.

The platform is designed with a user-friendly interface, making it accessible to a wide range of users, including both local and international tourists. In addition, by promoting attractions, restaurants, hotels, and local events, the system can contribute to the visibility of local tourism-related businesses and support the tourism industry.

Overall, the proposed system is expected to be well accepted by users due to its conve-

nience, ease of use, and ability to simplify the travel planning process.

### 2.5.2 Legal Feasibility

The proposed AI-powered Smart Tourism Assistant Platform is considered legally feasible, provided that it complies with applicable data protection regulations and the terms of use of third-party services. The system collects user information such as account details, travel preferences, and itineraries solely for providing personalized travel planning services. This information should be securely stored and used only for its intended purpose.

The platform also integrates with external services, including the Gemini API, Google Maps Platform, and Weather API. Therefore, the system must comply with the respective licensing agreements, usage policies, and Terms of Service of these providers. In addition, any tourism-related content, such as images or descriptions, should be obtained from authorized sources or used in accordance with applicable copyright and licensing requirements.

By following these legal and ethical considerations, no significant legal barriers have been identified that would prevent the successful implementation of the proposed system.

# 3 Considerations

Several important considerations must be addressed to ensure that the proposed AI-powered Smart Tourism Assistant Platform is practical, reliable, secure, and user-friendly. Since the system will support personalized trip planning through web and mobile applications, the most important concerns are performance, security, usability, data accuracy, maintainability, scalability, API dependency, and privacy.

## 3.1 Performance

Performance is an important consideration because users expect quick responses when generating itineraries, viewing nearby attractions, and accessing maps or weather information. The system should be designed to reduce unnecessary API calls and optimize database queries.Frequently accessed tourism data can be cached where appropriate to improve response time. The backend should also be structured to handle multiple requests efficiently.

## 3.2 Security

Security is essential because the system will store user accounts, preferences, and saved travel plans. User authentication should be implemented using JWT based authentication. Passwords should be securely hashed before storage. The system should also validate user inputs to prevent unauthorized access, injection attacks, and misuse of APIs. Admin functions should be protected so that only authorized administrators can add,update, or delete tourism related data.

## 3.3 Usability

The system should be simple and easy to use for different types of travelers. The conversational interface should allow users to enter travel requirements naturally without needing technical knowledge.The itinerary output should be clear, organized, and easy to understand. The user interface should support simple navigation, readable layouts, and mobile friendly design.

## 3.4 Maintainability

The system should be developed using a modular architecture. The frontend, backend, AI planning module, database, and administrative dashboard should be separated clearly. This makes the system easier to maintain, test, update, and extend in the future. New features such as booking links, user reviews, or advanced recommendation algorithms can be added later without changing the whole system.

## 3.5 Scalability

Although the first version will be developed as a prototype, the system should be designed to support future growth. If the number of users, destinations, attractions, and API requests increases, the system should be able to scale gradually. Docker based deployment and optimized backend services can support future expansion.

# References


[1] Meta Open Source, "React Documentation," [Online]. Available: https://react.dev/. [Accessed: Jul. 12, 2026].

[2] FastAPI, "FastAPI Documentation," [Online]. Available: https://fastapi.tiangolo.com/. [Accessed: Jul. 12, 2026].

[3] PostgreSQL Global Development Group, "PostgreSQL Documentation," [Online]. Available: https://www.postgresql.org/docs/. [Accessed: Jul. 12, 2026].

[4] SQLAlchemy, "SQLAlchemy Documentation," [Online]. Available: https://docs.sqlalchemy.org/. [Accessed: Jul. 12, 2026].

[5] Google AI for Developers, "Gemini Developer API Pricing," [Online]. Available: https://ai.google.dev/gemini-api/docs/pricing. [Accessed: Jul. 12, 2026].

[6] Google Maps Platform, "Google Maps Platform Core Services Pricing List," [Online]. Available: https://developers.google.com/maps/billing-and-pricing/pricing. [Accessed: Jul. 12, 2026].

[7] OpenWeather, "Self-Service API Pricing," [Online]. Available: https://openweathermap.org/price. [Accessed: Jul. 12, 2026].

[8] GitHub, "Pricing: Plans for Every Developer," [Online]. Available: https://github.com/pricing. [Accessed: Jul. 12, 2026].

[9] Vercel, "Vercel Pricing: Hobby, Pro, and Enterprise Plans," [Online]. Available: https://vercel.com/pricing. [Accessed: Jul. 12, 2026].

[10] Docker, "Docker Documentation," [Online]. Available: https://docs.docker.com/. [Accessed: Jul. 12, 2026].